# A DEADLY BACILLUS

**THE CAUSES, SYMPTOMS, DIAGNOSIS AND TREATMENT OF A LETHAL DISEASE THAT HAS ADAPTED TO EVADE CONVENTIONAL DRUGS**

Graphic by **TANMOY CHAKRABORTY;** Text by **SONALI ACHARJEE**

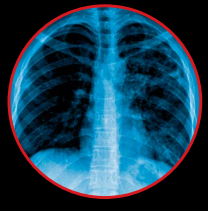
## WHAT IS TUBERCULOSIS?



It is a disease caused by the bacteria ***Mycobacterium tuberculosis***. Though it usually affects the lungs, it can also affect other parts of the body, such as the kidney, brain, liver or spine. However, not everyone who has the bacteria in their system develops TB. An infection that doesn't manifest as disease is called 'latent TB'



**Common symptoms** are persistent cough, blood in the sputum, unexplained weight loss, fever and unexplained chills. The disease spreads through the air, when an infected person coughs, speaks or sneezes. Pulmonary TB is the most contagious



When another person breathes in the bacteria, it settles in the lungs, where it can remain latent or multiply. **People with active TB can infect five to 15 people** in close contact within a year

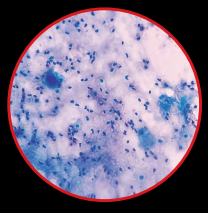
## WHEN DOES TB BECOME DRUG-RESISTANT?



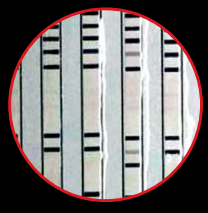

**About a quarter of TB patients** in India have resistance to one or the other TB drug. Wrong duration or dosage of antibiotics turns the bacteria resistant to drugs

MDR, **or Multi-Drug Resistant TB**, is resistant to isoniazid and rifampicin; **XDR or Extensively Drug-Resistant TB** to rifampicin, any fluoroquinolone, and at least one of the three injectible second-line drugs–amikacin, kanamycin or capreomycin

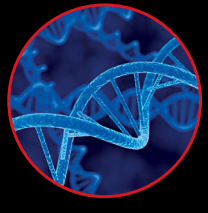
## HOW DO WE DETECT TB?



**Sputum Smear Microscopy:** It's the gold standard test to diagnose TB and uses sputum samples collected from the lungs. Since the illness is caused by slow-growing bacteria, it can take three to eight weeks to yield results, and even longer to confirm drug resistance



**Line-probe Assays:** The first molecular tests recommended by the WHO. They take an average of two days to diagnose TB and resistance to first- and second-line drugs. A fluid sample is taken from the patient's lungs and then tested on strips called 'assays'



**CBNAAT/TrueNat:** In this WHO-endorsed method, sputum or patient samples are checked using the highly specific, sensitive and relatively quicker nucleic-acid amplification tests. They can confirm not just TB but also resistance to rifampicin within two hours

## OTHER TYPES OF TB AND SYMPTOMS

*(TB in other parts of the body can exhibit diverse signs, which may or may not be accompanied by traditional symptoms like cough, fever or weight loss)*

**PLEURAL TB**
*(infection in pleura, the thin layer of tissue that covers the lungs and lines the interior wall of the chest cavity)*

- Breathlessness
- Chest pain
- Cough and weight loss

